

Deep Automotive Radar Super-Resolution Based on Array Extrapolation

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Abstract—Conventionally, the imaging quality of automotive radar is limited by the high sidelobes resulting from matched-filtering processing and by the coarse resolution caused by the limited number of radar channels. To overcome these limitations, we propose a novel deep learning-based automotive radar imaging algorithm incorporating array extrapolation. Specifically, an extrapolation mechanism for the echoes is developed based on an auto-regressive (AR) model, enabling data-scale extrapolation beyond the original aperture. Using the extrapolated echo data, a sparse-regularized imaging optimization model is then formulated. This problem is solved via an unfolded Approximate Message Passing (AMP) network, achieving learning-based super-resolution imaging. Finally, the effectiveness and superiority of the proposed method are validated through both simulations and real measured experiments, with comparisons to several classical imaging baselines.

Index Terms—Automotive Radar, Deep Unfolding, Extrapolation, Super-Resolution, Radar Imaging, Super-Resolution.

I. INTRODUCTION

Automotive radar [1] has become an essential sensing technology for advanced driver-assistance systems (ADAS) and autonomous driving [2], providing robust detection capabilities under adverse weather and lighting conditions. High-resolution radar imaging enables precise localization and characterization of multiple targets in complex traffic scenarios, which is critical for collision avoidance, path planning, and scene understanding. Conventional automotive radar imaging methods are primarily based on matched-filtering or back-projection algorithms [3], which convert the received echoes into spatial images of the target scene. However, these approaches face inherent limitations: matched-filtering often produces high sidelobes, leading to ambiguous target responses, while the spatial resolution is fundamentally constrained by the finite number of radar channels and the physical aperture of the antenna array. Consequently, improving the resolution and sidelobe suppression of automotive radar images has become a major research focus in recent years [4].

To overcome the resolution and sidelobe limitations of conventional radar imaging, sparse regularization-based techniques have been widely investigated [5]. These methods exploit the inherent sparsity of typical automotive scenes, where only a limited number of dominant scatterers exist relative to the full sensing grid, and formulate radar imaging as a sparse inverse problem [6]. Various optimization algorithms have been developed to solve these problems efficiently. For

example, iterative shrinkage-thresholding algorithms (ISTA) and its accelerated variants are commonly used for L1-norm regularized reconstruction [7]. The Alternating Direction Method of Multipliers (ADMM) has also been employed to handle complex constraints and improve convergence in sparse imaging [8], [9]. Furthermore, Approximate Message Passing (AMP) algorithms have been proposed as an efficient alternative [10], leveraging statistical properties of the measurement process to achieve fast convergence with reduced computational complexity. These sparse reconstruction methods have demonstrated significant improvements in spatial resolution and sidelobe suppression, particularly in scenarios with closely spaced targets or limited radar channels, establishing them as essential tools for high-resolution automotive radar imaging. Additionally, array extrapolation techniques have been studied to effectively enlarge the virtual aperture and enhance spatial resolution [11], [12]. These methods typically predict additional array measurements beyond the physically available elements, either by parametric modeling or signal subspace analysis. Auto-regressive (AR) models [13] and other linear prediction-based approaches have been widely used to extrapolate the array data [11], allowing finer angular sampling and narrower mainlobes. By increasing the effective aperture, array extrapolation can significantly improve the resolution of closely spaced targets and reduce the impact of limited channel numbers, making it a valuable complement to sparse reconstruction in high-resolution radar imaging.

However, conventional iterative optimization methods often suffer from high computational cost and slow convergence, limiting their practical applicability in real-time systems. To address these challenges, model-driven deep learning frameworks have been proposed, which unfold iterative optimization algorithms into trainable network architectures [14], [15]. By integrating the domain knowledge of the underlying physical and signal models, these networks retain interpretability while leveraging the learning capability of neural networks to accelerate convergence and enhance imaging performance.

Building on this idea, we propose a novel automotive radar imaging method that combines AR-based array extrapolation with a sparse-regularized optimization model solved via an unfolded Approximate Message Passing (AMP) network. First, an extrapolation mechanism based on an auto-regressive (AR) model is applied to the radar echoes, extending the data beyond the original aperture. Then, a sparse-regularized

imaging problem is formulated using the extrapolated data, and the unfolded AMP network is employed to achieve super-resolution, learning-based reconstruction. The effectiveness and superiority of the proposed approach are validated through both simulations and real measured experiments, showing significant improvements over classical imaging baselines.

II. PROBLEM FORMATION

We consider the automotive radar sensing scenario shown in Fig.1. Wherein, the red line corresponds to the radar antenna array mounted on the car. It is assumed that the antenna array is oriented along the y -axis, while the radar line-of-sight is aligned with the x -axis as shown in the figure. The position of measurement element is denoted by $\mathbf{p}_a = (0, y)$, while the coordinate of the scatterer is represented by $\mathbf{p}_t = (x, y)$.

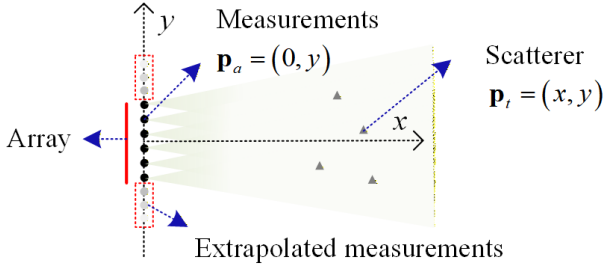


Fig. 1. Imaging geometry and the coordinate definitions.

We assume that the radar transmits a linear frequency modulated (LFM) signal with a pulse duration of T_p , and the transmitted signal model can be expressed as

$$s_t(t) = A_t \cdot \text{rect}\left(\frac{t}{T_p}\right) \exp(j(2\pi f_0 t + \pi K t^2)) \quad (1)$$

where A_t refers to the amplitude of the transmitted signal, K is the frequency slope, f_c stands for the carrier frequency, $\text{rect}(t)$ defines the rectangular window function.

We assume that the radar scene consists of Q point scatterers. The position of the m -th antenna measurement element is denoted as $(0, y_m)$, and the location of the q -th scatterer is given by (x_q, y_q) . The corresponding round-trip propagation distance between the measurement element and the scatterer can be expressed by

$$R_{m,q} = 2\sqrt{x_q^2 + (y_q - y_m)^2}. \quad (2)$$

Then, the received echo signal at the m -th measurement element can be modeled as the superposition of the echoes reflected by all scatterers, i.e.,

$$s_r^{(m)}(t) = \sum_{q=1}^Q \sigma_q s_t\left(t - \frac{R_{m,q}}{c}\right) \exp\left(-j\frac{2\pi f_c R_{m,q}}{c}\right), \quad (3)$$

where σ_q denotes the complex scattering coefficient of the q -th target and c is the speed of light.

After range compression and baseband demodulation, the received signal can be discretized with respect to fast time and antenna index, leading to a linear measurement model

for radar imaging. By stacking the received samples into a measurement vector $\mathbf{y} \in \mathbb{C}^M$, the imaging problem can be formulated as

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{n}, \quad (4)$$

where $\mathbf{x} \in \mathbb{C}^N$ represents the reflectivity coefficients of the discretized imaging grid, $\mathbf{A} \in \mathbb{C}^{M \times N}$ denotes the sensing matrix determined by the radar geometry and signal model, and $\mathbf{n}$ accounts for measurement noise. This linear inverse formulation provides the foundation for subsequent super-resolution automotive radar imaging.

III. THE PROPOSED METHOD

A. AR-Based Array Extrapolation

The imaging resolution of automotive radar is fundamentally limited by the physical aperture of the vehicle-mounted antenna array. In practical systems, the number of available antenna elements is restricted, leading to insufficient spatial sampling and degraded angular resolution. To mitigate this limitation, array extrapolation is employed to predict additional virtual measurements beyond the original aperture.

In our work, the radar echo is extrapolated along the array dimension using an auto-regressive (AR) model. For a fixed range cell, the received echoes across the antenna array are treated as a one-dimensional complex-valued spatial sequence. Let

$$\mathbf{y} = [y_1, y_2, \dots, y_M]^T \in \mathbb{C}^M \quad (5)$$

denote the measured echo vector corresponding to M antenna elements at a given range bin. This sequence is modeled as a P -th order AR process, expressed as

$$y_m = \sum_{p=1}^P a_p y_{m-p} + e_m, \quad m = P+1, \dots, M, \quad (6)$$

where $\{a_p\}_{p=1}^P$ are the AR coefficients and e_m represents the modeling error.

The AR coefficients are estimated from the observed echo data by solving the Yule-Walker equations. Specifically, let $r(k)$ denote the spatial autocorrelation of the echo sequence, defined as

$$r(k) = \mathbb{E}\{y_m y_{m-k}^*\}, \quad k = 0, 1, \dots, P. \quad (7)$$

The AR coefficients satisfy the following linear system:

$$\begin{bmatrix} r(0) & r(1) & \dots & r(P-1) \\ r(1)^* & r(0) & \dots & r(P-2) \\ \vdots & \vdots & \ddots & \vdots \\ r(P-1)^* & r(P-2)^* & \dots & r(0) \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_P \end{bmatrix} = \begin{bmatrix} r(1) \\ r(2) \\ \vdots \\ r(P) \end{bmatrix}. \quad (8)$$

Once the AR coefficients are obtained, forward extrapolation of the echo sequence can be performed as

$$\hat{y}_{M+n} = \sum_{p=1}^P a_p \hat{y}_{M+n-p}, \quad n = 1, \dots, N_{\text{ext}}, \quad (9)$$

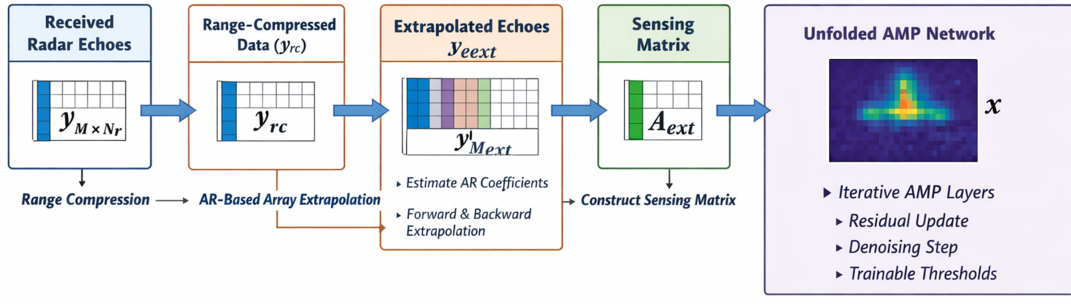


Fig. 2. Overview of the proposed automotive radar imaging pipeline. **Received Radar Echoes** ($\mathbf{y}$): Raw complex-valued signals measured by the vehicle-mounted antenna array for all range bins. **Range-Compressed Data** ($\mathbf{y}_{rc}$): Echoes after matched filtering in fast-time, improving range resolution. **Extrapolated Echoes** ($\mathbf{y}_{ext}$): Virtual array data obtained via AR-based array extrapolation, where the AR coefficients are estimated from the original array, and forward/backward prediction generates additional virtual elements. **Sensing Matrix** ($\mathbf{A}_{ext}$): Measurement matrix constructed according to array geometry and radar signal model, mapping the scene reflectivity to the extended echo data. **Unfolded AMP Network**: Iterative sparse reconstruction framework unfolded into a neural network structure, performing residual updates and soft-thresholding at each layer with trainable thresholds to reconstruct the high-resolution radar image $\mathbf{x}$. Arrows indicate the sequential processing flow from raw echoes to final high-resolution imaging.

and backward extrapolation can be carried out in a similar manner. By combining the extrapolated samples with the original measurements, an extended echo vector

$$\mathbf{y}_{ext} \in \mathbb{C}^{(M+2N_{ext})} \quad (10)$$

is constructed, where N_{ext} denotes the number of extrapolated elements on each side of the array.

By applying the AR-based extrapolation independently to each range cell, a virtual array with an enlarged effective aperture is formed, providing enhanced spatial sampling for subsequent high-resolution radar imaging.

B. Unfolded AMP Imaging for Automotive Radar

Based on the AR-extrapolated echo data, high-resolution radar imaging can be formulated as a sparse inverse problem. Let

$$\mathbf{y}_{ext} = \mathbf{A}_{ext}\mathbf{x} + \mathbf{n} \quad (11)$$

denote the automotive imaging measurement model, where $\mathbf{y}_{ext} \in \mathbb{C}^{M_{ext}}$ represents the extrapolated echo vector, $\mathbf{A}_{ext} \in \mathbb{C}^{M_{ext} \times N}$ is the sensing matrix determined by the radar geometry and signal model, $\mathbf{x} \in \mathbb{C}^N$ denotes the reflectivity coefficients of the discretized imaging grid, and $\mathbf{n}$ accounts for measurement noise.

The imaging problem is cast as an ℓ_1 -regularized optimization problem, expressed as

$$\min_{\mathbf{x}} \frac{1}{2} \|\mathbf{y}_{ext} - \mathbf{A}_{ext}\mathbf{x}\|_2^2 + \lambda \|\mathbf{x}\|_1, \quad (12)$$

where λ is the regularization parameter controlling the sparsity of the solution.

The Approximate Message Passing (AMP) algorithm provides an efficient iterative framework for solving sparse reconstruction problems of this form. Starting from an initial estimate $\mathbf{x}^0 = \mathbf{0}$, the AMP iterations are given by

$$\mathbf{r}^t = \mathbf{y}_{ext} - \mathbf{A}_{ext}\mathbf{x}^t + \frac{1}{\delta}\mathbf{r}^{t-1} \langle \eta'(\mathbf{x}^{t-1}) \rangle, \quad (13)$$

$$\mathbf{x}^{t+1} = \eta(\mathbf{x}^t + \mathbf{A}_{ext}^H \mathbf{r}^t; \theta^t), \quad (14)$$

where t refers to the iteration index, $\mathbf{r}^t$ denotes the residual vector, $\delta = M_{ext}/N$ is the measurement ratio, $\eta(\cdot)$ represents a soft-thresholding function, $\eta'(\cdot)$ denotes its derivative, θ^t is the iteration-dependent threshold, and $\langle \cdot \rangle$ denotes the empirical average over vector entries.

To improve convergence speed and reconstruction performance, the AMP iterations are unfolded into a finite-depth neural network, where each iteration corresponds to one network layer. In the unfolded AMP imaging network, the threshold parameters $\{\theta^t\}$ and other algorithmic parameters are treated as learnable variables and optimized through data-driven training. This model-driven design preserves the interpretability of the AMP algorithm while leveraging learning-based optimization to adapt to complex radar scenes and noise conditions.

By integrating the AR-based array extrapolation with the unfolded AMP imaging network, the proposed method achieves efficient super-resolution reconstruction for automotive radar imaging, effectively enhancing angular resolution and suppressing sidelobes under limited-channel radar sensing.

IV. RESULTS AND DISCUSSION

This section validates the effectiveness of the proposed algorithm through imaging experiments based on both simulated and real measured data. The simulation comparisons include MF and AMP. Furthermore, to verify the effectiveness of AR extrapolation, additional performance comparisons of AR+MF is presented.

In the simulations, a one-dimensional radar imaging scenario using a 64-channel radar array is considered. The radar operates at a carrier frequency of 77GHz, and the imaging region is discretized into 256 range bins. The target scene is assumed to be sparse and consists of three point targets located at different range positions with distinct reflection amplitudes. Complex additive white Gaussian noise is introduced to the received signals, with the noise power adjusted according to the specified signal-to-noise ratio (SNR). To comprehensively evaluate the performance of the proposed method under differ-

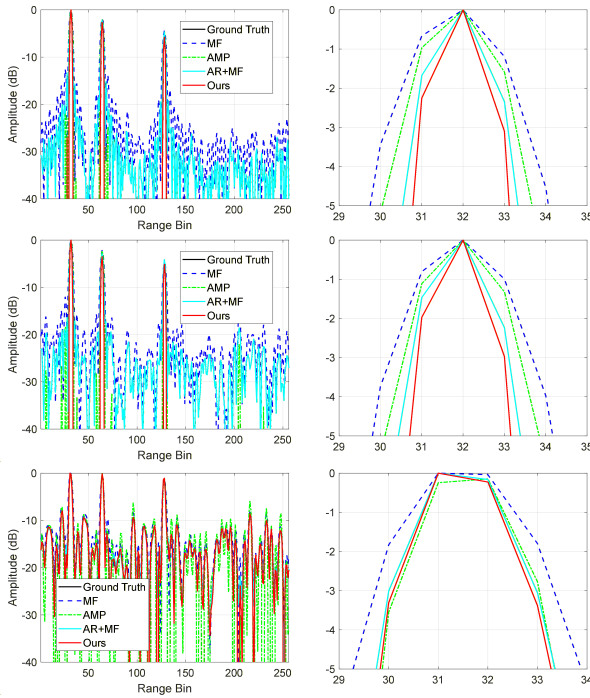


Fig. 3. Simulated range profiles of three point targets under different SNR conditions. Row 1-3 correspond to SNR being 20, 10 and 0 dB, respectively.

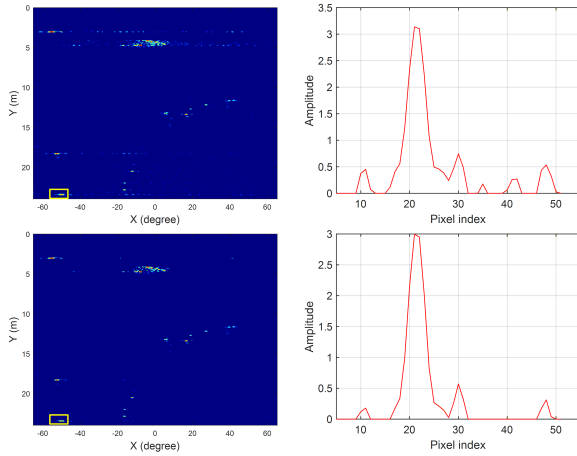


Fig. 4. Measured results of automotive array radar imaging, row 1 and 2 correspond to results by AMP and the proposed method, wherein, the right figures show the profiles of the strong scatterer circled in yellow in the left figures.

ent noise levels, imaging experiments are conducted at SNR = 20 dB, 10 dB, and 0 dB.

The imaging results are shown in Fig.3. It is observed from the figure that the proposed algorithm consistently outperforms the competing methods under all three noise conditions. A comparison between MF and AR+MF shows that the incorporation of the AR module effectively narrows the mainlobe, resulting in improved resolution. Overall, the proposed method achieves the narrowest mainlobe, as illustrated by the red curve, indicating the most pronounced resolution im-

provement, which validates the effectiveness of the proposed method.

In addition, real measured data are used to further validate the proposed algorithm. The data are collected using a TI 77-GHz cascaded millimeter-wave radar with 12 transmit antennas and 16 receive antennas. The radar is placed at a street corner to monitor pedestrian and vehicle targets within a range of 0–25 m. One frame of the measured data is selected for performance evaluation.

Fig.4 shows the imaging results by AMP and the proposed method. Wherein row 1 and 2 correspond to results by AMP and the proposed method, the right figures show the profiles of the strong scatterer circled in yellow in the left figures. From the figure, it is evident that the proposed algorithm achieves markedly reduced sidelobe levels. The target profile in the right figure further shows that the reconstructed strong scatterer exhibits a sharper and more pronounced peak than that of AMP, indicating an improvement in imaging resolution. The observed sidelobe reduction and resolution enhancement are mainly attributed to the combined effects of AR extrapolation and AMP deep unfolding. AR extrapolation effectively extends the radar aperture by predicting statistically consistent echo samples, which increases the spatial-frequency sampling density and leads to a more concentrated point spread function with a narrower mainlobe and lower sidelobes. Meanwhile, the unfolded AMP not only exploits the inherent sparsity of the scene through iterative shrinkage and residual correction, but also automatically learns optimal parameters within the unfolded layers, progressively suppressing noise and sidelobe artifacts while preserving dominant scatterers. The joint use of AR-based aperture extension and unfolded AMP with self-learned parameters thus yields sharper peaks for strong scatterers and significantly improved imaging resolution.

V. CONCLUSION

In this work, we present an automotive radar imaging algorithm that combines AR-based echo extrapolation with the unfolded AMP framework. By extending the received echoes beyond the original aperture and leveraging the sparsity of the scene, the proposed method effectively suppresses sidelobes and enhances imaging resolution. The deep unfolding mechanism further enables automatic learning of optimal parameters, improving reconstruction accuracy and robustness. Extensive simulations and real measured experiments demonstrate that the proposed approach consistently outperforms conventional MF and standard AMP methods, achieving sharper mainlobes and lower sidelobes. These results confirm the capability of the proposed algorithm to provide high-quality, super-resolution radar imaging even under limited-channel and noisy measurement conditions.

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